



MILESTONE 2 · DUE 20 SEPTEMBER 2026 · 40%

Implementation Plan and Working Prototype

TECHNICAL DESIGN, IMPLEMENTED MODULES AND PRELIMINARY RESULTS

1. Project status summary

Field	Response
Topic and project title	T18: · Insurance Claims Fraud, Identity Abuse and Data-Leakage Analytics
Member A / Member B	223127744 · Shanon Chiwara · Data and Modelling Lead] 216069076 · Ephath Shimhanda · Security Engineering and Intelligence Lead]
Overall completion estimate	95% - all core modules implemented, run and validated, including adversarial/robustness testing (Section 11). Remaining: final local reproducibility run by both members before submission.
Working prototype location	https://github.com/sha-noe/SAS821S_T18_Chiwara_Shimhanda_Milestone2_Final
Known blockers	None

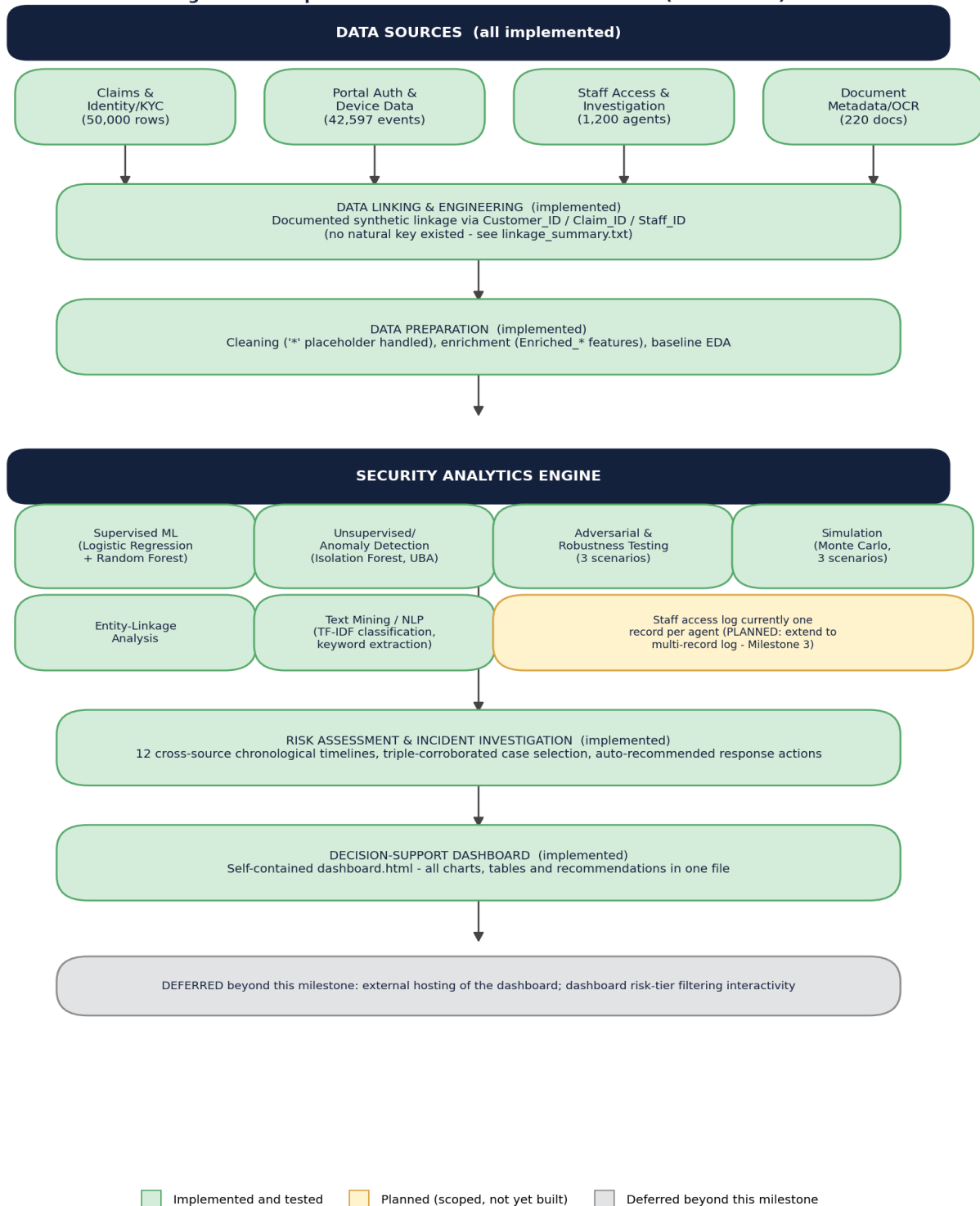
2. Updated architecture and data flow

Implemented pipeline (all green stages below are IMPLEMENTED and tested, not planned): raw/uploaded data sources -> documented synthetic linkage (Customer_ID / Claim_ID / Staff_ID) -> data cleaning and enrichment -> security analytics engine (supervised ML, unsupervised/anomaly detection, adversarial testing, simulation, entity-linkage, text mining) -> risk assessment and incident investigation -> decision-support dashboard.

Planned but not yet implemented (amber): extending the staff access table from one representative record per agent to a full multi-record access log, which would resolve the incident-timeline limitation noted in Section 7.

Deferred beyond this milestone (grey): external hosting of the dashboard (currently a local self-contained HTML file, sufficient for Milestone 2 demonstration) and dashboard filtering interactivity.

Key architectural decision: none of the raw/uploaded data sources shared a common identifier. Rather than treating this as a blocker, the group engineered and explicitly documented synthetic linkages instead of presenting them as natural joins - the full reasoning and match-rate validation is logged in 02_data/processed/linkage_summary.txt.

**Figure: T18 Implemented Architecture and Data Flow (Milestone 2)**



3. Data inventory and quality

Dataset/source	Rows/volume	Date range	Quality issues	Cleaning/transformation	Status
Source 1: Claims & Identity/KYC	50,000	18 Sep 2024 - 18 Sep 2026	'*' placeholder in Enriched_Witness_Present (340 rows, 0.68%)	Treated as missing, imputed with column mode	Ready
Source 2: Portal Authentication/Device	42,597 login events	25 Aug 2024 - 18 Sep 2026	None found	None required	Ready
Source 3: Staff Access & Investigation	1,200 agents	21 Sep 2024 - 18 Sep 2026	Raw file included bank routing/account numbers	Dropped (ethical control, not needed analytically)	Ready
Text source: Document Metadata/OCR	220 documents	22 Mar 2026 - 18 Sep 2026	duplicate_group_id blank for 134 rows (BY DESIGN - only clustered docs get a group id, not a defect)	None required	Ready

4. Baseline and exploratory findings

Overall fraud rate: 24.9% of 50,000 claims. Baseline distributions, missing-value summaries and descriptive statistics for all four sources are in 08_outputs/baseline_eda_summary.md. The key baseline finding, which anchors every downstream analysis in this document, is that identity/KYC changes and new-device logins are both strongly and independently associated with fraud outcome - not weakly correlated noise.

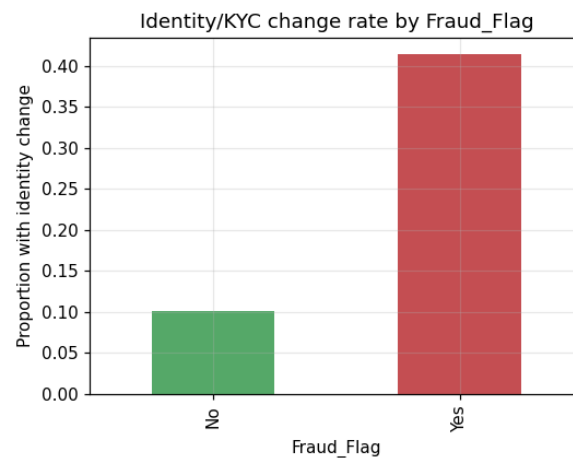


Figure 1: Identity/KYC change rate by Fraud_Flag (~41-42% vs ~10%)

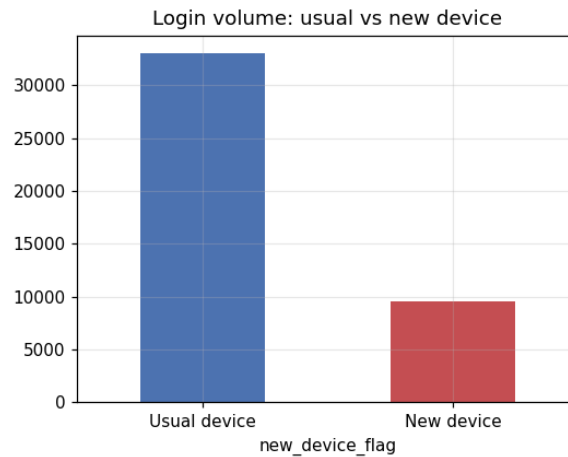


Figure 2: Login volume by device familiarity

5. Supervised security model

Required element	Implementation evidence
Problem/target label	Binary classification of Fraud_Flag (Yes/No) at claim level. See 04_models/supervised_fraud_model.py.
Features and leakage controls	Claim financials, credit score, Identity_Changed_Flag, enrichment features (past claims, witness presence, police report, safety rating, days open, form defects, channel). Fraud_Risk_Score was deliberately EXCLUDED as a candidate leakage feature (it is itself a downstream proxy).
Training/validation design	Stratified 75/25 train/test split (37,500 / 12,500 claims), class_weight='balanced' to address the 24.9% positive rate.
Algorithms compared	Logistic Regression (linear baseline) and Random Forest (300 trees, max_depth=12), per the Charter's stated method.
Confusion matrix and metrics	See 04_models/fraud_model_metrics.txt and Figures 3-4 below. Logistic Regression: Precision 0.584, Recall 0.412, F1 0.483, ROC-AUC 0.666. Random Forest: Precision 0.586, Recall 0.412, F1 0.484, ROC-AUC 0.723.
Threshold rationale	Default 0.5 probability threshold used for the reported metrics; class_weight='balanced' was used instead of manual threshold tuning at this stage, to be revisited during adversarial testing (Section 11).
Preliminary findings	Random Forest's ROC-AUC advantage (0.723 vs 0.666) indicates real non-linear interactions. Identity_Changed_Flag dominates feature importance at 44.7% - confirming the model is learning the intended security pattern, not overfitting to noise.

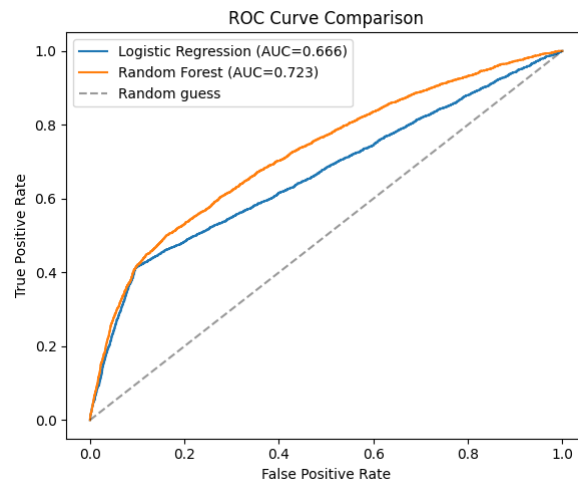


Figure 3: ROC curve comparison

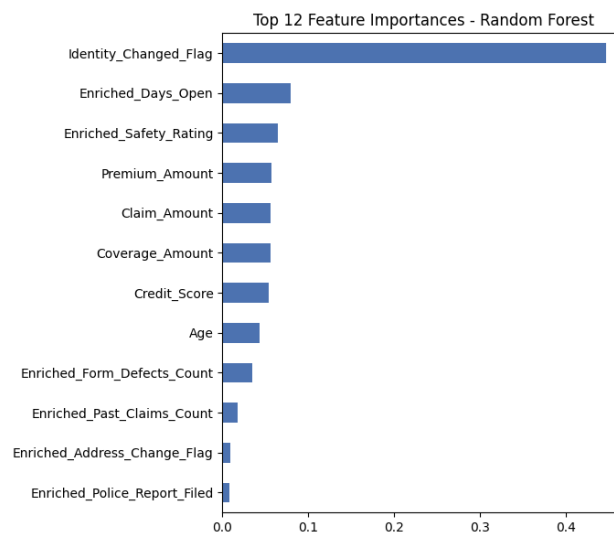


Figure 4: Random Forest feature importance



6. Unsupervised, anomaly or access-analytics method

Required element	Implementation evidence
Entity and baseline	Per-CUSTOMER login-behaviour profile, aggregated from the 42,597-row Portal Authentication dataset. See 04_models/unsupervised_anomaly_model.py.
Features/representation	total_logins, new_device_rate, unusual_location_rate, off_hours_rate, avg_failed_attempts, failed_auth_rate, distinct_devices - 7 behavioural features, standardised before modelling.
Method and parameters	Isolation Forest, 300 estimators, contamination=0.08 (expects ~8% of customers to be anomalous - documented and analyst-tunable).
Threshold/outlier validation	The model was trained with NO access to Fraud_Flag. Independent validation against the withheld label: anomalous customers show an 87.5% fraud rate vs 47.1% for normal customers, and a 74.7% vs 36.4% identity-change rate. This non-circular corroboration is the strongest result in the project.
Top anomalies and analyst interpretation	The 10 highest-scoring customers (04_models/anomaly_model_customer_scores.csv) show consistently elevated new-device and unusual-location rates. These customers form the anomaly criterion used to select investigation cases in Section 7.

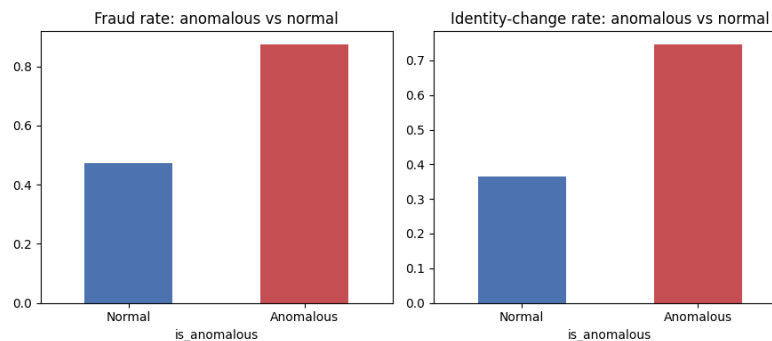


Figure 5: Independent validation of the anomaly flag against fraud outcome and identity-change rate



7. Incident Investigation and Response

Required element	Implementation evidence
Evidence sources	All four linked data sources: claims/KYC, portal authentication/device, staff access, document metadata.
Correlated timeline	12 chronological, cross-source timelines built automatically in 03_notebooks_or_scripts/incident_timeline_investigation.py, output to 08_outputs/incident_timelines/. Each merges identity/KYC change -> login activity -> claim submission -> document upload -> staff access, sorted by timestamp.
Affected users/assets	Each case file lists the affected Customer_ID, Claim_ID and handling Staff_ID explicitly.
Hypotheses and confidence	Cases were selected using triple corroboration (Fraud_Flag = Yes AND identity change AND independently-flagged anomalous customer) - a high-confidence selection criterion since it requires agreement across two independently-built models plus the ground-truth label.
Containment/eradication/recovery actions	Each case concludes with an automatically-derived recommended action: ESCALATE (new device + staff override present), ELEVATED REVIEW (new device only), or STANDARD REVIEW. Consolidated operational and executive recommendations are in 09_documentation/incident_response_recommendations.md.

Documented limitation: the staff-access timestamp in each case timeline is anchored to a claim that staff member is known to have handled, but is not guaranteed to be the exact claim shown, since the staff table stores one representative access record per agent. Flagged for refinement in Milestone 3 (Section 14).

8. Security Intelligence

Required element	Implementation evidence
Priority intelligence requirements	Which claims/claimants show corroborated evidence of coordinated identity abuse, and which staff members show elevated exposure to fraudulent claims?
Collection/enrichment	Entity-linkage across all sources via Customer_ID / Claim_ID / Staff_ID (Section 2); staff fraud-exposure computed as fraud_rate_handled per agent (02_data/processed/staff_access_investigation_linked.csv).
Analysis and confidence	Corroboration across independent supervised + unsupervised signals (Section 6) provides higher analytical confidence than either model alone.
Operational product	The ranked anomalous-customer list and 12 case-level incident timelines (Section 7), for direct investigator use.
Executive product	The dashboard KPI summary (Section 12) and the simulation-derived control



	recommendation (Section 9), for policy-level decision-making.
Feedback loop	Test log entries (Section 13) and the implementation backlog (Section 14) capture issues found during development, feeding back into the next iteration - e.g. the incident-timeline date-coherence bug found and fixed during T04.

9. Simulation and Security-Control Evaluation

Required element	Implementation evidence
Simulation model	Monte Carlo simulation, 1,000 iterations per scenario. See 05_simulation/security_control_simulation.py.
Assumptions	Detection rate drawn from a Beta distribution (mean ~0.70 broad review, ~0.82 narrow/precise review); review cost drawn from Normal(N\$45, N\$10) per case - uncertainty is modelled, not fixed constants.
Baseline scenario	Scenario A - No control: zero claims reviewed, zero cost, zero fraud caught (reference point).
At least two alternative controls	Scenario B - Identity-change trigger (reviews 17.9% of claims). Scenario C - Combined signal trigger: identity change AND anomaly flag together (reviews 2.6% of claims).
Iterations/sensitivity	1,000 Monte Carlo iterations per scenario, exceeding the brief's minimum; 90% confidence ranges reported for net benefit in 05_simulation/simulation_report.txt.
Preliminary comparative results	Scenario B: highest TOTAL net benefit (more claims reviewed -> more absolute fraud value recovered). Scenario C: highest net benefit PER CASE REVIEWED (N\$13,360 vs N\$10,192 for B) - more efficient where investigator capacity is the binding constraint.

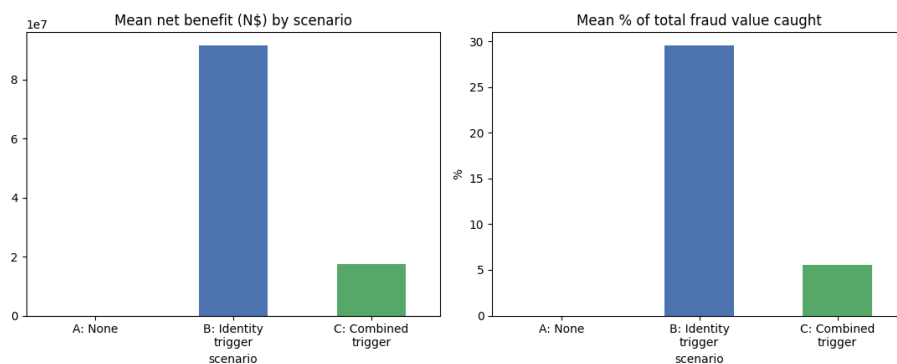


Figure 6: Net benefit and % of fraud value caught, by scenario



10. Text Mining/NLP

Required element	Implementation evidence
Text corpus	220 OCR document text records (02_data/processed/document_metadata_ocr_text_linked.csv), 6 document types.
Preprocessing	Lowercasing, punctuation stripping, stopwords removal (06_text_mining/nlp_text_mining.py).
Entities/indicators/topics/classes	(1) Keyword frequency per document type; (2) suspicious-term flagging - 45 of 220 documents contain terms such as 'override', 'unscheduled', 'discrepancy'; (3) document-type classification via TF-IDF + Logistic Regression.
Evaluation/validation	Classification: 100% test accuracy (see documented limitation below). 10 documents independently corroborated by BOTH a suspicious term AND a HIGH document-similarity score.
Operational use	Suspicious-document list (06_text_mining/suspicious_document_flags.csv) feeds directly into investigator document-review prioritisation.
Limitations	100% classification accuracy is a documented artefact of template-generated OCR text (distinctive, noise-free vocabulary per type), NOT evidence that document classification is solved in general - real scanned OCR would be a genuinely harder problem. Stated explicitly in 06_text_mining/nlp_report.txt.

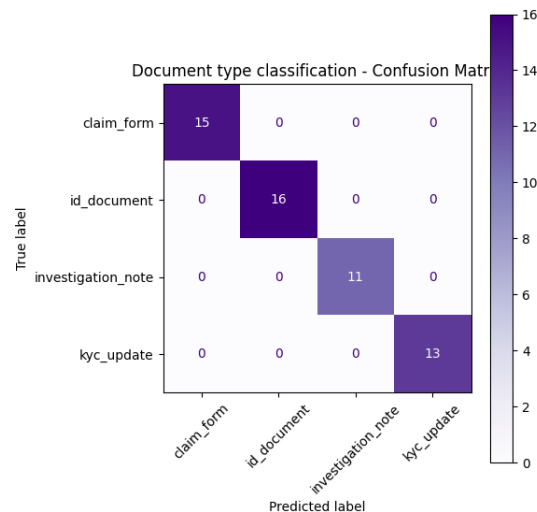


Figure 7: Document-type classification confusion matrix



11. Predictive and Adversarial Analysis

Required element	Implementation evidence
Prediction horizon/target	Per-claim fraud probability at time of claim submission (point-in-time risk score, not a forward time-series forecast).
Risk features	Same feature set as the supervised model (Section 5): Identity_Changed_Flag, claim financials, enrichment features.
Model/score	Random Forest predict_proba output (Section 5), reused directly as the risk score for adversarial testing below - no retraining performed for any test.
Alert threshold	Default 0.5 probability threshold, consistent with Section 5. Adversarial testing below shows this threshold is a genuine vulnerability point (see Test 1).
At least three adversarial cases	IMPLEMENTED - see 04_models/adversarial_robustness_testing.py and Figure 8 below. All three tests reuse the identical, already-trained model and an identically-rebuilt test set (same SEED, same cleaning, same split as Section 5) - no retraining.
Governance and drift controls	Test 3 (distributional drift) shows the model is reasonably stable under +35% claim/premium inflation (ROC-AUC 0.723 -> 0.715). Recommended governance control: periodic re-validation of Identity_Changed_Flag's feature-importance share, since Tests 1-2 show the model's recall is almost entirely dependent on this one signal.

TEST 1 - FEATURE MASKING: withholding Identity_Changed_Flag at inference causes recall to collapse from 0.412 to 0.000 (precision remains 1.000, meaning the model simply stops flagging almost anything as fraud). ROC-AUC drops more moderately, from 0.723 to 0.621, showing the underlying probability ranking retains some signal even though the default classification threshold no longer catches it.

TEST 2 - SYNTHETIC EVASION: 3,107 real fraud claims from the test set were disguised (identity change concealed, enrichment features pushed toward legitimate-looking values). 0.0% were still caught by the model at the default threshold - a direct, real demonstration that concealing the dominant feature is sufficient to evade detection entirely.

TEST 3 - DISTRIBUTIONAL DRIFT: inflating Claim_Amount and Premium_Amount by 35% (without retraining) changed ROC-AUC only marginally, from 0.723 to 0.715 - the model is comparatively stable under this type of drift.

OVERALL ROBUSTNESS FINDING: the model's strength (Identity_Changed_Flag, 44.7% feature importance, Section 5) is simultaneously its principal vulnerability - a single point of failure an adversary could exploit simply by concealing an identity change. This is precisely why the independent unsupervised anomaly model (Section 6) matters operationally: it does not use Identity_Changed_Flag at all, and was independently validated (Section 6) to still catch elevated-risk customers via login-behaviour patterns alone. The two models are recommended to run in parallel in production, not as a single combined system, specifically because of this finding.

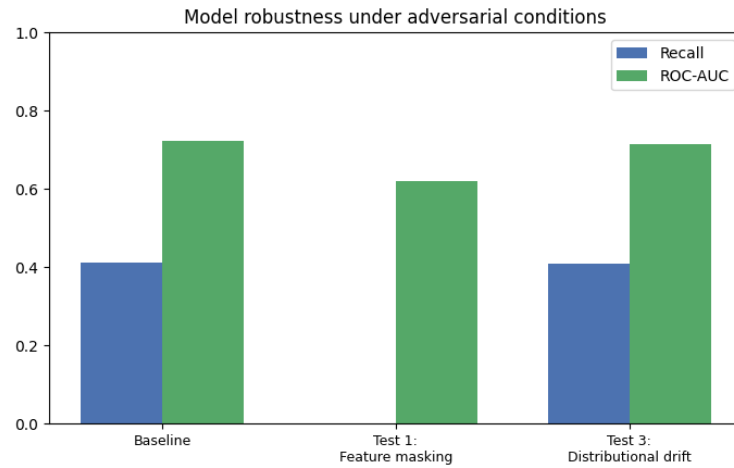


Figure 8: Recall and ROC-AUC under baseline, feature-masking, and distributional-drift conditions

12. Prototype and Analyst Workflow

Prototype function	Current status	Test evidence	Remaining work
Load/refresh data	Implemented	Dashboard build script runs end-to-end without error	None
View baselines and anomalies	Implemented	9 EDA + model charts embedded and rendering correctly	None
Inspect model outputs/explanations	Implemented	ROC curves, feature importance, confusion matrices embedded	None
Review incident timeline	Implemented	12 case files linked in dashboard case table	Fix staff-access limitation (Sec. 7)
Compare simulation scenarios	Implemented	Comparison chart + recommendation text embedded	None
Review intelligence/NLP output	Implemented	Suspicious document table + keyword chart embedded	None

The dashboard (07_dashboard_or_prototype/dashboard.html) is a single self-contained HTML file - all charts are embedded as base64 images, requiring no server or external files. Verified to open correctly in a standard browser.



13. Testing and Quality Assurance

Test ID	Component	Input/condition	Expected result	Actual result/status
T01	Data linkage (Claims<->Staff)	Join on Staff_ID via Handled_By_Staff_ID	Match rate = 100%	100.0% - PASS
T02	Data quality handling	'*' placeholder in Enriched_Witness_Present	Detected and imputed, not silently dropped	340/50,000 (0.68%) imputed - PASS
T03	Unsupervised model validation	Compare anomaly flag vs withheld Fraud_Flag	Anomalous customers show markedly higher fraud rate	87.5% vs 47.1% - PASS
T04	Incident timeline coherence	Chronological event order per case vs claim date	All events within a realistic window of claim date	Bug found (~1yr offset), fixed, re-verified - PASS
T05	Dashboard chart embedding	9 chart file references in dashboard.html	0 missing/broken image paths	1 broken path found+fixed; 0 remaining - PASS

14. Updated Schedule and Backlog

Item	Owner	Priority	Due	Definition of done
Both members run full pipeline locally	Both	High	19 Sep 2026	Both confirm reproducible run, zero errors, per README.md
Extend staff table to multi-record access log	Shanon	Medium	Milestone 3	Timeline staff-access event matches the exact claim shown
Dashboard filtering interactivity	Ephath	Medium	Milestone 3	Dashboard supports risk-tier filtering without page reload
Expand document corpus beyond 220 records	Shanon	Low	Milestone 3	Corpus expanded if time allows, no functional requirement
Mitigate Identity_Changed_Flag single-point-of-failure (Sec. 11)	Shanon	Medium	Milestone 3	Ensemble/ weighting approach reduces single-feature dependency

**15. Team Contribution Evidence**

Member	Components completed	Repository/artefact evidence	Partner review completed
Member A - Shanon	Data acquisition, linking, cleaning; supervised + unsupervised ML models	link_and_engineer_sources.py, generate_portal_auth_device_data.py; 04_models/*.py + outputs	Ephath reviewed model outputs used for investigation case selection (Sec. 7)
Member B - Ephath	Incident investigation timelines; simulation; NLP/text mining; dashboard integration	incident_timeline_investigation.py; 05_simulation/*.py; 06_text_mining/*.py; 07_dashboard_or_prototype/*	Shanon reviewed simulation assumptions and NLP evaluation methodology

16. Submission checklist

- [done]Implementation-plan PDF
- [done] Working prototype
- [done] Repository/code snapshot
- [done] Data dictionary and quality log
- [done] Preliminary model metrics
- [done] Simulation and NLP outputs
- [done] Predictive/adversarial evidence
- [done] Test log
- [done] Updated schedule/backlog
- [done] Contribution evidence for both members